LAMMPS official dem11-regionwall

Curved-wall rebound and geometry refinement | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Rebound vx (m/s)	-1 relative limit 0.01	-0.9998074 err 0.0001926 PASS	-0.9998074 err 0.0001926 PASS	-0.9998818 err 0.0001182 PASS	-0.9998818 err 0.0001182 PASS

Check time=0.000205 s; original step 20500. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
sphere	R=0.0025 rho=7000	(0.0072, 0, 0)	(2, 0, 0)	(0, 0, 0)

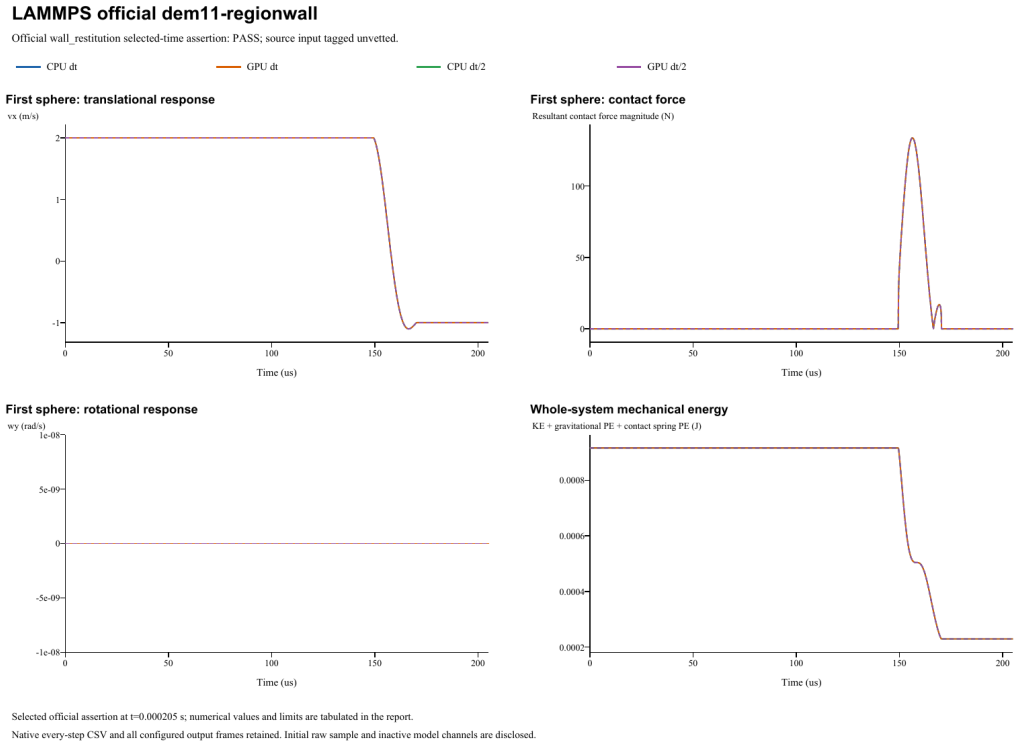
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=5.70157227e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 20500/41000 steps. The physical cylinder radius is 0.01 m with endcaps at z=+/-0.05 m. Sphere pose, velocity, mass, restitution and duration are retained; periodic Z is inactive at z=vz=0. The analytic conversion E_eff=(3/4)*kn*sqrt(8/7) maps the source inner-cylinder effective radius Rp*8/7 to native Rp, preserving elastic and automatic damping coefficients without fitting.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob e90c87ee07bed0d26ab25bb04320a43f73ef1b4c. Official tag: unvetted. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Rebound vx (m/s)	0.00019257	0.00011816	relative: 0.01

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 4.44e-16 m/s / 0 rad/s / 2.84e-14 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 6.94e-18 kg m/s / 4.33e-19 m / 2.2e-16 m/s. Gravity is added separately to saved native contact forces.

Geometry check	Grid h (m)	Axis SDF error (um)	Restitution
Fine CPU main	0.00025	1.0417437	0.4999037149
Coarse CPU diagnostic	0.0005	4.1679848	0.4998392524

Four fine-grid main runs retain 824 frames; the coarse CPU diagnostic adds 206, total 1030 frames in five executions. Actual-contact axis error improves 4.00097 times. E_eff=5.70157227e+10 Pa analytically preserves cylinder curvature; the prepared grid phase is part of the saved input. The source URL correction is metadata only.

Complete results and verified scope

All 1,030 native frames are retained; main frame counts=[206, 206, 206, 206]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.